Audio Deepfake Detection Using Traditional Speech Features and Machine Learning

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***Abstract*—**The rapid development of speech synthesis and voice conversion technologies has increased the difficulty of distinguishing genuine human speech from artificially generated or manipulated speech. This work investigates traditional speech features combined with machine-learning classifiers for audio deepfake detection using a balanced subset derived from the ASVspoof 2021 Speech Deepfake (DF) data. The common experimental protocol uses fixed training, validation, and testing partitions. The completed **MFCC** experiment represents each recording using the mean and standard deviation of 20 MFCC coefficients, producing a 40-dimensional vector. Linear and radial basis function (RBF) support vector machines are evaluated. The best MFCC configuration, an RBF-SVM with C=10 and gamma=0.1, achieves 94.64% accuracy, 94.91% precision, 94.34% recall, 94.62% F1-score, 98.90% ROC-AUC, and 5.30% EER on the held-out test subset. The completed **LFCC** experiment represents each recording using the mean and standard deviation of 13 LFCC coefficients, producing a 26-dimensional vector. The best LFCC configuration, an RBF-SVM with C=10 and gamma=0.1, achieves 86.57% accuracy, 87.06% precision, 85.90% recall, 86.48% F1-score, 93.93% ROC-AUC, and 13.36% EER on the held-out test subset. On the validation set, the LFCC-based RBF-SVM achieves 87.47% accuracy, 88.49% precision, 86.14% recall, 87.30% F1-score, and 94.58% ROC-AUC. These results demonstrate the effectiveness of traditional LFCC features combined with an RBF-SVM for speech deepfake detection. For the **spectrogram-based** branch, a compact three-block convolutional neural network (28,034 parameters) trained directly on log-mel spectrograms achieves 84.53% accuracy, 81.18% precision, 89.88% recall, 85.31% F1-score, 92.50% ROC-AUC, and 17.00% EER on the held-out test subset. Across all three branches, the MFCC-based RBF-SVM achieves the strongest performance on every metric, followed by the LFCC-based RBF-SVM, with the CNN weakest overall — though its recall exceeds LFCC's.

***Index Terms*—**audio deepfake detection, ASVspoof, MFCC, LFCC, spectral features, SVM, speech processing.

I. Introduction

Recent advances in text-to-speech synthesis and voice conversion can generate increasingly realistic speech. These technologies are useful in many applications, but they can also be used to create audio deepfakes that imitate genuine human speech. Audio deepfake detection therefore aims to identify acoustic characteristics or artifacts that distinguish bonafide speech from spoofed speech. This project studies traditional speech-processing representations under a common machine-learning protocol. The team investigates MFCC, LFCC, and spectral feature representations, with the goal of comparing their ability to separate bonafide and spoof recordings while avoiding data leakage and using consistent evaluation metrics.

II. Dataset and Experimental Setup

A. Dataset

The experiments use audio from ASVspoof 2021 Speech Deepfake (DF), specifically Part00. Part00 contains 152,955 FLAC files: 147,420 spoof and 5,535 bonafide samples. Because this distribution is highly imbalanced, all 5,535 bonafide samples were retained and 5,535 spoof samples were randomly selected with random_state=42, producing a balanced experimental subset of 11,070 recordings. Labels are encoded as bonafide=0 and spoof=1, with spoof treated as the positive class.

B. Fixed Data Partitioning

The same stratified train/validation/test partitions are used across feature experiments to support a fair comparison. The training set is used for learning model parameters, the validation set is used for model and hyperparameter selection, and the test set remains held out until final evaluation.

Split	Bonafide	Spoof	Total
Training	3,874	3,875	7,749
Validation	830	830	1,660
Testing	831	830	1,661
Total	5,535	5,535	11,070

C. Audio Preprocessing

For the MFCC experiment, recordings are processed as mono audio at 16 kHz and peak normalized. The analysis uses an FFT size of 512, a hop length of 256 samples, a Hann window, centered framing, and high-quality resampling (soxr_hq) when required.

III. Feature Extraction

A. MFCC

Twenty Mel-Frequency Cepstral Coefficients (MFCCs) are extracted using 40 Mel filters, DCT type 2, orthonormal DCT normalization, FFT size 512, and hop length 256. Because recordings have different durations, temporal statistical pooling converts each variable-length MFCC sequence to a fixed vector. For every coefficient, its mean

and standard deviation are computed, yielding 20 means + 20 standard deviations = 40 features per recording. Feature extraction completed for all 11,070 recordings with zero extraction failures.

B. LFCC

Thirteen Linear Frequency Cepstral Coefficients (LFCCs) are extracted using **40 linear filters**, with an FFT size of **512** and a hop length of **256**. Statistical pooling computes the **mean and standard deviation of each coefficient**, resulting in **13 mean + 13 standard deviation values = 26 features per recording**. These fixed-length LFCC features are then used as input to the SVM classifier.

C. Spectrogram

Log-mel spectrograms are extracted from mono audio resampled to 16 kHz, using 128 Mel filters, an FFT size of 2048, and a hop length of 512 samples. Unlike the MFCC and LFCC branches, which apply temporal statistical pooling to obtain a fixed-length vector, the spectrogram representation retains the time-frequency structure of each recording: every clip is fixed to a 4-second duration (padding shorter clips and truncating longer ones), producing a 128 × 126 log-mel matrix per recording. Each spectrogram is standardized per clip before being passed to the convolutional classifier [confirm this description matches your *extract_melspectrogram* function exactly]. Feature extraction completed for all 11,070 recordings with zero extraction failures.

IV. Classification Methodology

A. Standardization

MFCC features are standardized with StandardScaler. The scaler is fitted only on the training features and then applied to training, validation, and test data. This prevents validation or test distribution statistics from leaking into model training.

B. SVM Model Selection

Linear and RBF SVM kernels are evaluated. The regularization parameter is searched over $C \in \{0.1, 1, 10, 100\}$. For RBF models, gamma $\in$ {scale, 0.01, 0.1, 1}. Validation F1-score is used as the primary selection criterion. The selected MFCC configuration is RBF-SVM with C=10 and gamma=0.1.

C. CNN Architecture

The spectrogram branch uses a lightweight three-block 2D CNN. Each block applies a 3×3 convolution (16, 32, and 64 filters across the three blocks), followed by batch normalization, ReLU activation, and 2×2 max pooling. Global average pooling follows the final block, feeding a 64-unit dense layer (0.3 dropout) and a two-unit softmax output. The network totals 28,034 parameters (27,810 trainable).

Training runs for up to 40 epochs with Adam (initial learning rate 1×10^{-3}, batch size 32). Early stopping (patience 7, restoring best weights) and learning-rate reduction on plateau both monitor validation loss; the lowest-validation-loss checkpoint, not the final epoch, is evaluated. Since the training set is nearly balanced (3,874 bonafide vs. 3,875 spoof), class weighting has negligible effect. A grid search over learning rate ($\{1\times10^{-3}, 1\times10^{-4}\}$) and dropout ({0.2, 0.4}) was attempted but did not finish in time, so the configuration above was used directly rather than a search-selected one.

V. Evaluation Metrics

Performance is reported using accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix, and Equal Error Rate (EER). Precision, recall, and F1 use spoof as the positive class. ROC-AUC summarizes discrimination across decision thresholds, while EER is obtained by linear interpolation of the empirical ROC at the point where FPR = 1 − TPR.

VI. Results and Discussion

A. MFCC Validation and Test Results

Metric	Validation	Test
Accuracy	95.48%	94.64%
Precision	94.67%	94.91%
Recall	96.39%	94.34%
F1-score	95.52%	94.62%
ROC-AUC	99.03%	98.90%
EER	—	5.30%

The best validation configuration uses the RBF kernel with C=10 and gamma=0.1. The held-out test results remain close to the validation results: test F1 is 94.62% compared with 95.52% on validation, and test ROC-AUC is 98.90% compared with 99.03% on validation. The linear SVM configurations achieved validation F1 scores around 80%, whereas the best RBF configuration reached 95.52%. This suggests that a nonlinear decision boundary is more suitable for the selected 40-dimensional MFCC representation.

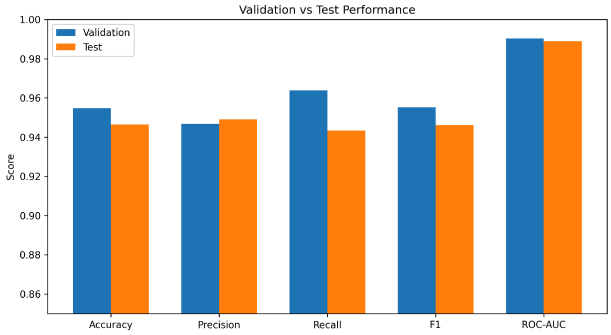


Fig. 1. Validation vs. test performance for the MFCC-based RBF-SVM.

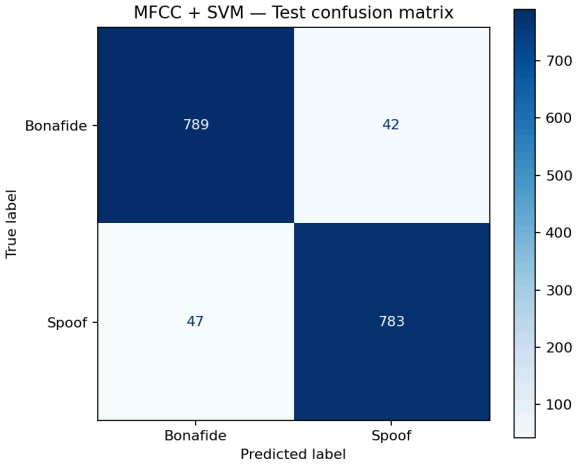


Fig. 2. Confusion matrix of the MFCC-based RBF-SVM on the held-out test subset.

The confusion matrix contains 789 correctly classified bonafide recordings and 783 correctly classified spoof recordings. Forty-two bonafide samples were classified as spoof, and 47 spoof samples were classified as bonafide. Overall, 1,572 of 1,661 test recordings were correctly classified, with 89 errors.

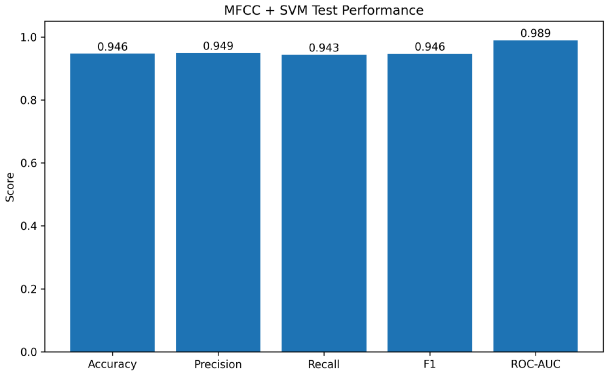


Fig. 3. Final MFCC + RBF-SVM test performance.

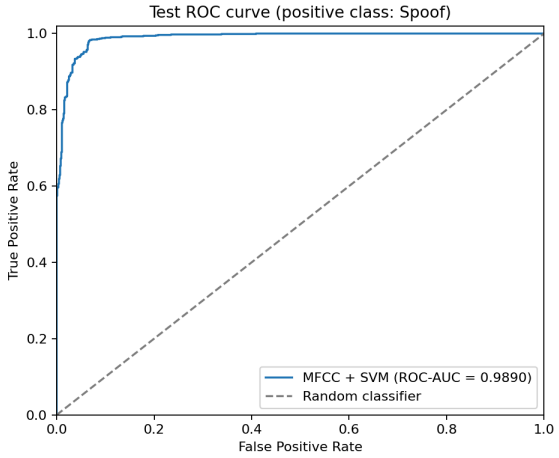


Fig. 4. Test ROC curve for the MFCC-based RBF-SVM (positive class: Spoof).

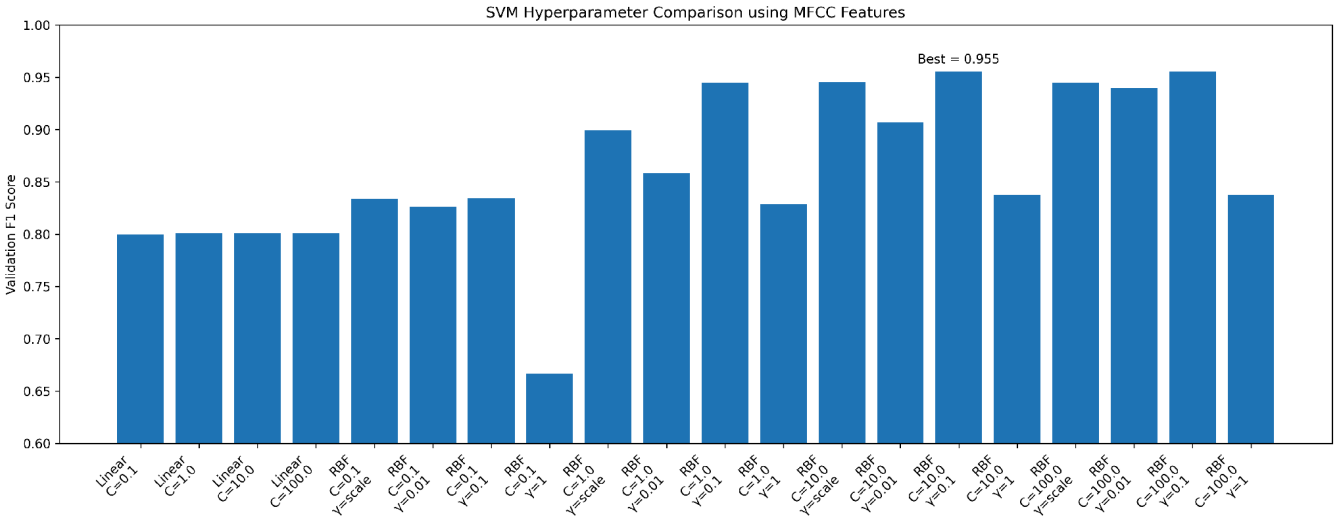


Fig. 5. Validation F1-score across the evaluated MFCC-SVM hyperparameter configurations.

B. LFCC Validation and Test Results

Metric	Validation	Test
Accuracy	87.47%	86.57%

Metric	Validation	Test
Precision	88.49%	87.06%
Recall	86.14%	85.90%
F1-score	87.30%	86.48%
ROC-AUC	94.58%	93.93%
EER	-	13.36%

The best validation configuration for the LFCC-based system uses the RBF kernel with C = 10 and gamma = 0.1. The held-out test results remain close to the validation results: test F1 is 86.48% compared with 87.30% on validation, and test accuracy is 86.57% compared with 87.47% on validation. The Linear SVM achieved a validation F1-score of 75.04%, whereas the best RBF configuration achieved 87.30%. This indicates that the relationship between the LFCC-based features and the speech classes is better captured by a non-linear decision boundary.

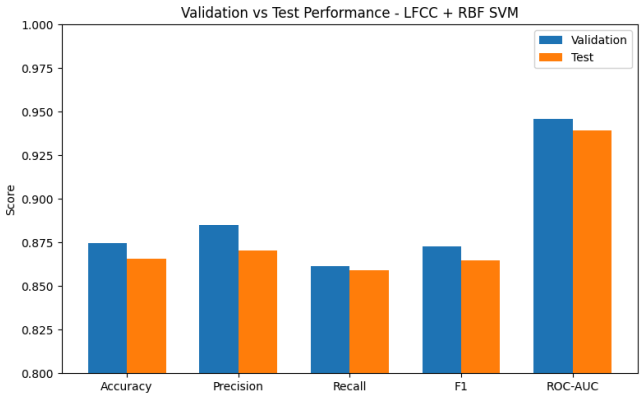


Fig 6 .Validation vs. test performance for the LFCC-based RBF-SVM.

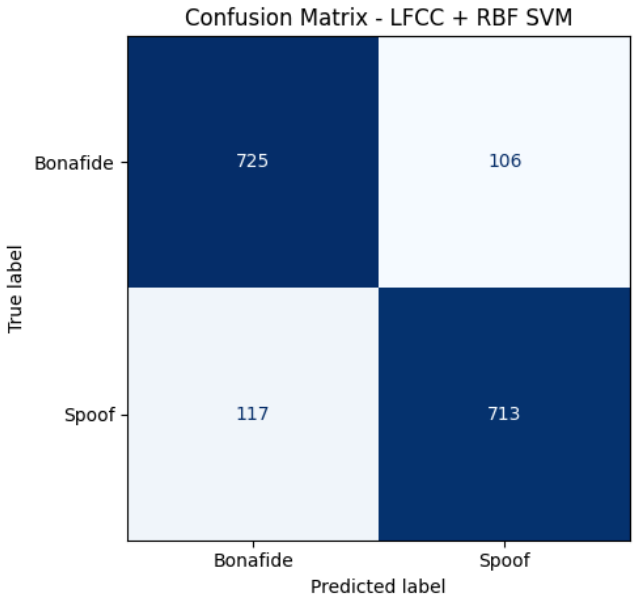


Fig 7 . Confusion matrix of the LFCC-based RBF-SVM on the held-out test subset.

The confusion matrix contains 725 correctly classified bonafide recordings and 713 correctly classified spoof recordings. One hundred six bonafide samples were classified as spoof, and 117 spoof samples were classified as bonafide. Overall, 1,438 of 1,661 test recordings were correctly classified, with 223 errors.

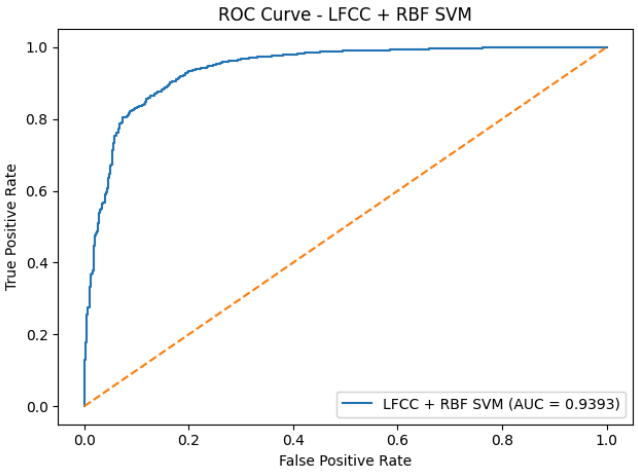


Fig 8 . Test ROC curve for the LFCC-based RBF-SVM (positive class: Spoof).

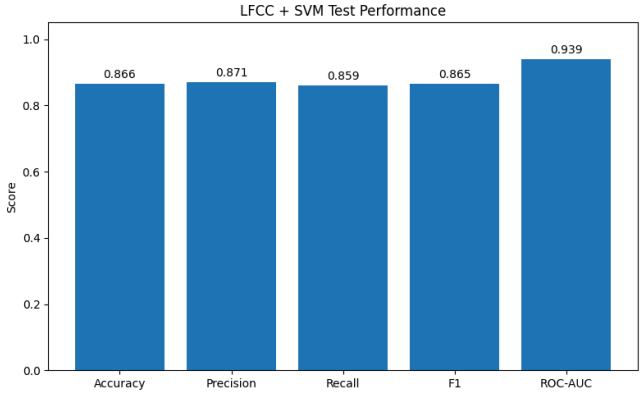


Fig 9 . Final LFCC + RBF-SVM test performance.

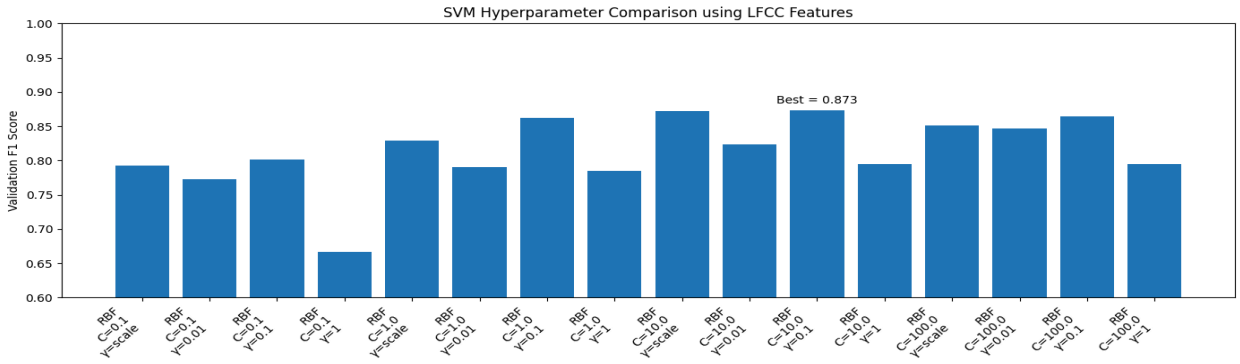


Fig 10 . Validation F1-score across the evaluated LFCC-SVM hyperparameter configurations.

C.Spectrogram + CNN Results and Discussion

Metric	Test
Accuracy	84.53%
Precision	81.18%
Recall	89.88%
F1-score	85.31%
ROC-AUC	92.50%
EER	17.00%

The CNN is trained directly on log-mel spectrograms [*Fig 11.real vs. fake mel-spectrogram comparison*] rather than a hand-crafted statistical summary. Its 84.53% test accuracy and 17.00% EER are the weakest of the three branches on most metrics — below both LFCC (86.57%, 13.36% EER) and MFCC (94.64%, 5.30% EER) — though its 89.88% recall exceeds LFCC's 85.90%, both trailing MFCC's 94.34% [Fig. 13]. The confusion matrix [Fig. 14] contains 658 correctly classified bonafide recordings and 746 correctly classified spoof recordings; 173 bonafide samples were classified as spoof and 84 spoof samples as bonafide, giving 1,404 of 1,661 test recordings correct (257 errors).

Training again showed substantial epoch-to-epoch fluctuation in validation loss and accuracy [Fig. 12] — validation accuracy ranged from roughly 0.50 to 0.83 across 15 epochs while training accuracy rose smoothly to about 0.85. Retaining the lowest-validation-loss checkpoint (around epoch 7) rather than the final epoch's weights kept the reported test metrics tied to the model's best observed point during this run.

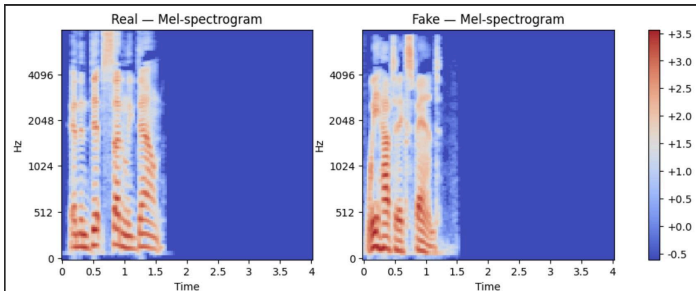


Fig 11 . real vs. fake mel-spectrogram comparison

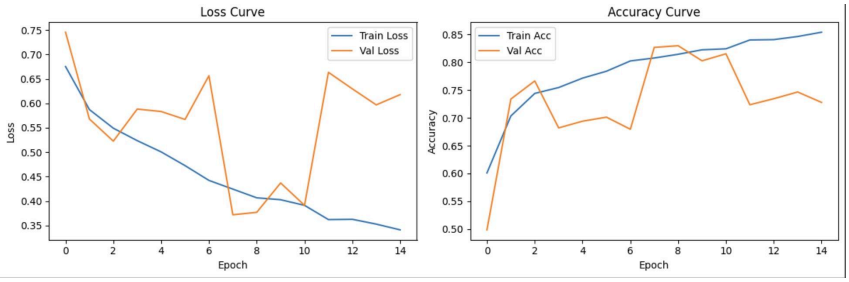


Fig 12 loss/accuracy curves

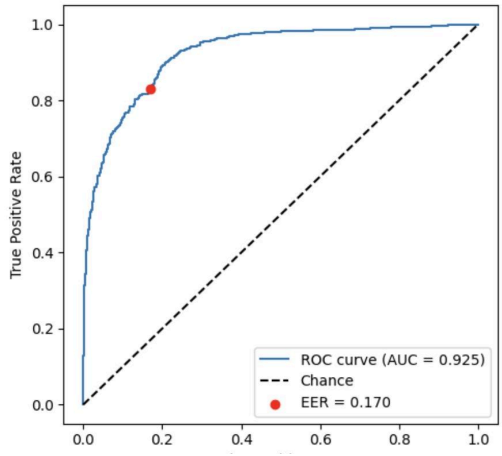


Fig 13 ROC curve

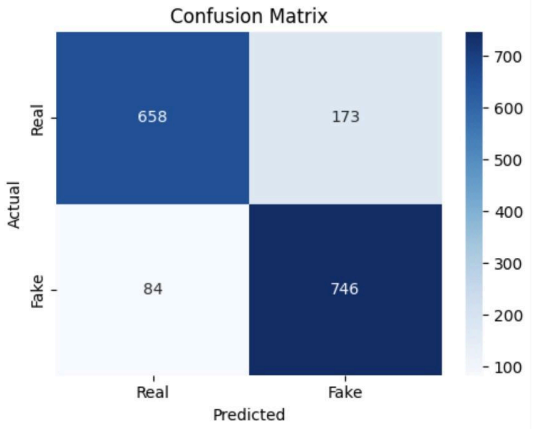


Fig 14 Confusion matrix

VII. Comparative Analysis

Feature	Classifier	Accuracy	Precision	Recall	F1	ROC-AUC	EER
MFCC	RBF-SVM	94.64%	94.91%	94.34%	94.62%	98.90%	5.30%
LFCC	RBF-SVM	86.57%	87.06%	85.90%	86.48%	93.93%	13.36%
Spectrogram	CNN	84.53%	81.18%	89.88%	85.31%	92.50%	17.00%

MFCC-based RBF-SVM achieves the strongest performance on every metric. The spectrogram-based CNN is weakest overall of the three branches, though its recall (89.88%) exceeds the LFCC-based system's (85.90%) — correctly flagging a larger share of spoofed clips, at the cost of more false positives on bonafide speech than either SVM branch. A plausible explanation for the SVM branches' advantage is that mean/standard-deviation pooling over cepstral coefficients yields a compact, well-separated representation that an RBF kernel — exhaustively tuned over 16 hyperparameter combinations — can exploit effectively even with a modest training set. The CNN, by contrast, learns features directly from raw time-frequency data with no statistical pooling, and was tuned far less (2 of 4 planned configurations completed); its weaker performance likely reflects this smaller search budget rather than a shortcoming of spectrogram features in principle.

VIII. Limitations and Generalization

The reported MFCC results are obtained on a balanced subset constructed from Part00 of ASVspoof 2021 DF rather than the complete official ASVspoof 2021 evaluation protocol. Therefore, the reported 94.64% accuracy and related metrics describe performance on this study's held-out test subset and should not be presented as results on the complete official DF evaluation set. In addition, training, validation, and testing samples originate from the selected ASVspoof source. Cross-dataset evaluation on unseen synthesis and voice-conversion systems would provide stronger evidence of generalization. Finally, mean and standard-deviation pooling provides a compact representation but discards temporal ordering in the MFCC sequence, The CNN also proved sensitive to training run: two runs under identical code, data, and configuration produced test accuracies of 89.95% and 84.53% respectively, depending on which epoch's validation loss was lowest when early stopping triggered. Because Colab's

local runtime storage does not persist across sessions, only the second run's checkpoint remained available for final evaluation — underscoring that a single run does not reliably characterize this architecture's performance here, and that persisting checkpoints immediately after training matters in ephemeral cloud environments.

IX. Conclusion

This study investigates traditional speech features with machine-learning classifiers for audio deepfake detection. Under the completed MFCC experiment, 20 MFCC coefficients pooled by temporal mean and standard deviation form a 40-dimensional representation. The selected RBF-SVM (C=10, gamma=0.1) achieves 94.64% accuracy, 94.62% F1-score, 98.90% ROC-AUC, and 5.30% EER on the held-out test subset.Under the LFCC experiment, 13 LFCC coefficients pooled by temporal mean and standard deviation form a 26-dimensional representation. The selected RBF-SVM (C=10, gamma=0.1) achieves 86.57% accuracy, 86.48% F1-score, 93.93% ROC-AUC, and 13.36% EER on the held-out test subset Under the spectrogram + CNN experiment, log-mel spectrograms (128 Mel filters, 4-second fixed duration) are used directly as input to a compact three-block convolutional neural network (28,034 parameters), avoiding the temporal pooling used in the MFCC and LFCC branches. The trained CNN achieves 84.53% accuracy, 85.31% F1-score, 92.50% ROC-AUC, and 17.00% EER on the held-out test subset — the weakest of the three branches overall, though its recall (89.88%) exceeds LFCC's.

AI Tool Disclosure

Generative AI tools were used to assist with code organization, debugging, documentation, and language editing. Experimental results were obtained by executing the implemented signal-processing and machine-learning pipeline on the selected dataset and were not generated by the AI tool.

References

[1] ASVspoof 2021 / Speech Deepfake dataset reference